

Quiz 1: The Sun

Partial Answer Key

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Question 1

What is the wavelength, in nm, of a photon with energy of 1,240 eV?

To solve for wavelength of a photon with a given energy:

$$E = \frac{hc}{\lambda} \rightarrow E[\text{eV}] = \frac{1240}{\lambda[\text{nm}]} \rightarrow \lambda[\text{nm}] = \frac{1240}{E[\text{eV}]}$$

here is the solution

$$\frac{1240\text{eV}}{1240\text{eV}} = 1 \text{ nm}$$

Question 3

Assume the photosphere rotates $11.5^\circ/\text{day}$ at a latitude of 60°N and $14.3^\circ/\text{day}$ at the equator. How many more rotations per year will occur at the equator than at 60°N ? (1 rotation = 360°)

At lat = 60°N :

$$\frac{11.5^\circ}{\text{day}} \times \frac{365 \text{ days}}{\text{year}} \times \frac{1 \text{ rotation}}{360^\circ} \simeq 11.7 \text{ rotations/year}$$

At the equator:

$$\frac{14.3^\circ}{\text{day}} \times \frac{365 \text{ days}}{\text{year}} \times \frac{1 \text{ rotation}}{360^\circ} \simeq 14.5 \text{ rotations/year}$$

The difference is ~ 2.8 rotations per year.

Question 11

For a brown dwarf star with an approximate blackbody with a temperature of 2500 K, at what wavelength (in nm) does the peak in the radiation spectrum occur?

Like in HW1, use Wien's Displacement Law:

$$\lambda = \frac{\alpha}{T} = \frac{2.898 \times 10^{-3} \text{ m K}}{2500 \text{ K}} = 1160 \text{ nm}$$

This is a wavelength in the **infrared**; but the distribution has a tail that extends into the red part of the spectrum, so these stars appear reddish.

Question 14

Exposure to the sun's ultraviolet (UV) radiation in space can lead to increased spacecraft temperatures. Why?

You might think the increased UV exposure leads to more energy absorbed and therefore higher temperature. But remember that UV is only a tiny fraction of the solar energy output; most of the 1361 W/m² is in the visible and near-infrared.

Instead, what we're looking for here is the effect that solar UV has on materials and their surface properties. UV exposure can lead to an increase in the absorptivity of a material, and that change in the material property leads to increased temperature through the equations discussed in class.

We gave credit for any answer that discussed changes in the spacecraft material properties and thus changes to absorptivity (or emissivity).

Question 16

Suppose the nominal heliopause is at 100 AU, with a solar wind velocity of 400 km/s. Now, the solar wind velocity increases to 565.7 km/s, but the heliopause location doesn't change. Explain how this could happen. What else must change, and by how much? (You should be able to answer in just 2-3 sentences, but you can elaborate as much as you like.) [hint: refer to the equations you used in HW1, problem 5.]

In HW1, we found the following eq. for heliopause position:

$$\begin{aligned} P_{ISM} &= P_{SW} = \rho v^2 = m_p n v^2 \\ \frac{n_E}{n_{hp}} &= \frac{r_{hp}^2}{r_E^2} \rightarrow n_{hp} = n_E \frac{r_E^2}{r_{hp}^2} \\ P_{ISM} &= m_p n_E \frac{r_E^2}{r_{hp}^2} v^2 \rightarrow r_{hp} = \sqrt{\frac{m_p n_E r_E^2 v^2}{P_{ISM}}} \end{aligned}$$

Proton mass (m_p) and position of the Earth cannot be changed. The solar wind density must decrease by half or the external pressure of the interstellar medium must increase by a factor of two:

$$\begin{aligned} \sqrt{n_1} v_1 &= \sqrt{n_2} v_2 \rightarrow n_2 = \frac{v_1^2}{v_2^2} n_1 = \frac{(400)^2}{(565.7)^2} n_1 \simeq 0.5 n_1 \\ v_1 / \sqrt{P_1} &= v_2 / \sqrt{P_2} \rightarrow P_2 = \frac{v_2^2}{v_1^2} P_1 = \frac{(565.7)^2}{(400)^2} P_1 \simeq 2 P_1 \end{aligned}$$

Some people wrote this second answer, that the interstellar pressure would double to match the solar wind dynamics pressure! That wasn't an answer I was really looking for, but technically you're right, so we gave credit for that. (But what are the odds that the interstellar pressure just happens to double at the same time??)

Question 17

Assume the sun is losing mass through solar wind at a rate of 10⁹ kg/s and that the solar wind is flowing at 425 km/s. How much solar wind kinetic energy per second would be intercepted by the Earth's magnetosphere at 1 AU, assuming a magnetosphere radius of 15 R_E ? Submit your answer in J/s (Watts).

$$\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{1}{2} \frac{dm}{dt} v^2 = \frac{1}{2} (10^9 \text{ kg/s}) (4.25 \times 10^5 \text{ m/s})^2 = 9 \times 10^{19} \text{ J/s}$$

$$(9 \times 10^{19} \text{ J/s}) \times \left(\frac{1}{4\pi(1 \text{ AU})^2} \right) \times (\pi(15R_E)^2) \sim 9 \times 10^{12} \text{ J/s}$$

So about 10^{13} Watts, or 10,000 GW. That might sound like a lot, but it's a tiny, tiny fraction of the solar energy output: $1361 \text{ W/m}^2 \times (\pi(15R_E)^2)$ is 1.74×10^{17} Watts! So the solar wind energy flux is about 0.01% of the total solar output.

Question 18

A $5 \times 10^{13} \text{ kg}$ CME travelling at 2500 km/s just after it left the sun accelerated about 3×10^{34} 35-MeV particles into interplanetary space (this is an SEP event!). Consider the total kinetic energy of this CME. What percentage of the total kinetic energy of the CME went into accelerating these 35-MeV protons?

$$\frac{KE_{particles}}{KE_{CME}} = \frac{(3 \times 10^{34})(35 \text{ MeV})}{\frac{1}{2} m_{CME} v_{CME}^2} = \frac{(3 \times 10^{34})(5.607 \times 10^{-12} \text{ J})}{\frac{1}{2} (5 \times 10^{13} \text{ kg})(2.5 \times 10^6 \text{ m/s})^2} \simeq 0.001 = 0.1\%$$

Again, the particle output is a tiny fraction of the CME output, which in turn is a tiny fraction of the solar energy output...

Lest you start thinking all this stuff is so tiny that it really doesn't matter, rest assured we'll be seeing the effects on the magnetosphere and on spacecraft soon!

Question 19

Consider a cube-shaped spacecraft in Earth orbit, 50 cm on each side, with one side directly facing the sun. The spacecraft structure is covered in solar cells, which have absorptivity = 0.9 and emissivity = 0.7. The spacecraft generates and uses 22 W of power internally (using, for example, a radioisotope thermoelectric generator RTG); this internally-generated power is converted to heat. What is the equilibrium temperature of the spacecraft, in Kelvins?

As discussed in the Module 1 Course Notes, in Section 4.1:

$$Q_{in} + 22 \text{ W} = Q_{out} \rightarrow \alpha A_n S + 22 \text{ W} = \epsilon A_{tot} \sigma T^4$$

$$T = \left(\frac{\alpha S A_n + 22 \text{ W}}{\epsilon \sigma A_{tot}} \right)^{1/4} = \left(\frac{0.9(1361 \text{ W/m}^2)(0.5 \text{ m})^2 + 22 \text{ W}}{0.7(5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4})6(0.5 \text{ m})^2} \right)^{1/4} = 272.7 \text{ K}$$